Linear Molecules in the Aliev Framework

Implemented equations, optical constant L, and the final general decompacted branch

Implementation and derivation note

2026-03-26

Abstract

This note records the current linear-molecule implementation of the Aliev framework, including the optical scalar L, the Gaussian bootstrap used to construct the input layer, and the derivation of the final general linear observable from the Aliev–Watson formalism. The main conclusion is that the compact 1986 off-diagonal parallel formulas are not sufficient as a universal final observable representation. The structurally correct final branch is the decompacted linear observable, in which genuinely mode-anchored terms remain in one-mode coefficients while the unresolved parallel off-diagonal sector is carried explicitly by unordered pair contributions. The compact 1986 branch is therefore retained only as a conditional legacy compression, admissible when the pair-anchored sector is negligible.

Reader guide. Sections 1–4 fix notation and the implemented primary families. Sections 5–8 state the explicit linear coefficients and the observable state dependence. Sections 9–10 collect the benchmark evidence and the derivation of the parallel off-diagonal branch. Section 11 states the final implementation status and the paper-level conclusion. Nothing has been removed from the earlier audit material; this version only reorganizes it into a manuscript-style note.

1. Scope

This note is a standalone record of the current linear-molecule implementation in `/Users/vincenzobarone/centrifugal/linear_dv_aliev_terms.py` and of the Gaussian bootstrap in `/Users/vincenzobarone/centrifugal/scripts/build_linear_aliev_payload_from_gaussian.py`.

The aim is limited and explicit:

- write all implemented linear Aliev equations in a clean form;
- include the quartic optical scalar L;
- separate what is implemented algebraically from what is only frozen as a diagnostic convention.

This is not a claim that every remaining convention has been derived from first principles. It is the cleanest complete statement of the current linearmolecule branch.

Interpretive proposition. The present document should be read as a rigorous implementation audit of the linear Aliev branch together with the statement of a corrected working branch. It is not claimed to be a fully closed derivation of the original formalism in every remaining convention-dependent point.

2. Indices and objects

- Parallel modes are indexed by $n, n', n'', \ldots$.
- Perpendicular degenerate pairs are indexed by $t, t', \ldots$.
- B is the equilibrium rotational constant.
- D_J is the equilibrium quartic centrifugal distortion constant.
- ω_n and ω_t are harmonic frequencies of parallel and perpendicular modes.
- B_n^{xx} is the parallel rotational derivative entering $C_n = -BB_n^{xx}/\omega_n$.
- k'_{ijk} and k'_{ijkl} denote the cubic and quartic constants in the Aliev-side coordinate convention used by the implementation.

3. Primary definitions

The implemented primary objects are

$$C_n = -\frac{BB_n^{xx}}{\omega_n}, \tag{1}$$

$$X^{nt} = \frac{2B\zeta_{nt}\sqrt{\omega_n\omega_t}}{\omega_n^2 - \omega_t^2}, \tag{2}$$

$$X_{nt} = \frac{B\zeta_{nt}(\omega_n^2 + \omega_t^2)}{\sqrt{\omega_n\omega_t}(\omega_n^2 - \omega_t^2)}, \tag{3}$$

$$r_n = 4\omega_n C_n \left(\frac{D_J}{B} - \frac{B^2}{\omega_n^2}\right) + \frac{1}{2}\sum_{n',n''} k'_{nn'n''} C_{n'} C_{n''}, \tag{4}$$

$$r_{nn'} = \frac{3\omega_n\omega_{n'} C_n C_{n'}}{4B} + \frac{1}{2}\sum_{n''} k'_{nn'n''} C_{n''}, \tag{5}$$

$$r_{tt'} = \frac{1}{2}\sum_{n} k'_{tt'n} C_n. \tag{6}$$

4. Auxiliary families

The implemented F, U, and V families are

$$F^{nn'} = B^2 \sum_t \zeta_{nt}\zeta_{n't}\sqrt{\omega_n\omega_{n'}}\frac{\omega_n^2 + \omega_{n'}^2 - 2\omega_t^2}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \tag{7}$$

$$F_{nn'} = r_{nn'} + B^2 \sum_t \zeta_{nt}\zeta_{n't}\frac{\omega_n^2\omega_{n'}^2 - \omega_t^4}{\omega_t\sqrt{\omega_n\omega_{n'}}(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \tag{8}$$

$$U_{nn'} = \frac{r_{nn'}}{4(\omega_n + \omega_{n'})} + \frac{B^2}{8}\sum_t \frac{\zeta_{nt}\zeta_{n't}}{\omega_t(\omega_n\omega_{n'})^{1/2}} \frac{(\omega_n + \omega_t)^2(\omega_{n'} - \omega_t)^2 + (\omega_n - \omega_t)^2(\omega_{n'} + \omega_t)^2}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \tag{9}$$

$$V_{nn'} = \frac{r_{nn'}}{4(\omega_n - \omega_{n'})} + \frac{B^2}{8}\sum_t \frac{\zeta_{nt}\zeta_{n't}}{\omega_t(\omega_n\omega_{n'})^{1/2}} \frac{(\omega_n + \omega_t)^2(\omega_{n'} + \omega_t)^2(\omega_n + \omega_{n'} - 2\omega_t) - (\omega_n - \omega_t)^2(\omega_{n'} - \omega_t)^2(\omega_n + \omega}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)(\omega_n - \omega_{n'})} \tag{10}$$

For the perpendicular branch the implementation also keeps the analogous families

$$U_{tt'} = \frac{r_{tt'}}{4(\omega_t + \omega_{t'})} + \frac{B^2}{8}\sum_n \frac{\zeta_{nt}\zeta_{nt'}}{\omega_n(\omega_t\omega_{t'})^{1/2}} \frac{(\omega_t+\omega_n)^2(\omega_{t'}-\omega_n)^2 + (\omega_t-\omega_n)^2(\omega_{t'}+\omega_n)^2}{(\omega_t^2-\omega_n^2)(\omega_{t'}^2-\omega_n^2)}, \tag{11}$$

$$V_{tt'} = \frac{r_{tt'}}{4(\omega_t - \omega_{t'})} + \frac{B^2}{8}\sum_n \frac{\zeta_{nt}\zeta_{nt'}}{\omega_n(\omega_t\omega_{t'})^{1/2}} \frac{(\omega_t+\omega_n)^2(\omega_{t'}+\omega_n)^2(\omega_t+\omega_{t'}-2\omega_n) - (\omega_t-\omega_n)^2(\omega_{t'}-\omega_n)^2(\omega_t+\omega_{t'}+}{(\omega_t^2-\omega_n^2)(\omega_{t'}^2-\omega_n^2)(\omega_t-\omega_{t'})} \tag{12}$$

The text in the source scan explicitly licenses the substitution $n, n' \mapsto t, t'$ only for the F families. For $U_{tt'}$ and $V_{tt'}$ the present note therefore uses a controlled inference: one applies the same formal replacement rule directly to the implemented $U_{nn'}$ and $V_{nn'}$ kernels, namely

$$n, n' \mapsto t, t', \qquad \sum_t \mapsto \sum_n .$$

Under that substitution, the $U_{tt'}$ formula written above is reproduced exactly, and the corresponding $V_{tt'}$ branch takes the form written above. A direct structural check then shows that this $V_{tt'}$ is antisymmetric under $t \leftrightarrow t'$, as expected from the explicit factor $(\omega_t - \omega_{t'})$ in the denominator. So, in the present note, $U_{tt'}$ and $V_{tt'}$ are not claimed as explicitly printed Aliev formulas, but as the unique formulas obtained by the same formal $nn' \to tt'$ rule that the source text states explicitly for the F families.

5. Parallel coefficient β_n

The implemented parallel coefficient is

$$\begin{aligned}
\beta_n = {} & 4\omega_n^2 C_n^2\left(\frac{B}{\omega_n^2} - \frac{D_J}{B^2}\right) + \frac{1}{4}\sum_{n',n''} k^{(4)}_{\beta_n}(n; n', n'')C_{n'}C_{n''} - \sum_{n'} \frac{k'_{nnnn'}}{2\omega_{n'}} r_{n'} \\
& - 4B\sum_{n',t} C_{n'}\zeta_{n't}\frac{(3\omega_n^2+\omega_t^2)\omega_n^{3/2}C_n\zeta_{nt} + (\omega_n^2+\omega_t^2)\omega_n^{3/2}C_n r_{nn'}}{\sqrt{\omega_n\omega_t}(\omega_n^2-\omega_t^2)} \\
& + 4\sum_{n',t}\left(X_{nt}X_{n't}F^{nn'} + X^{nt}X^{n't}F_{nn'}\right) - 4\sum_{n',t}\left(X_{nt}X^{n't}F_{n'n} + X^{nt}X_{n't}F^{n'n}\right) \\
& - 16\sum_{n'}\left[U_{nn'}(\omega_n+\omega_{n'}) + V_{nn'}(\omega_{n'}-\omega_n)\right].
\end{aligned} \tag{13}$$

Implementation-side decomposition. The implemented β_n is organized as the sum of seven structurally distinct blocks:

$$\begin{aligned}
\beta_n = {} & \underbrace{4\omega_n^2 C_n^2\left(\frac{B}{\omega_n^2} - \frac{D_J}{B^2}\right)}_{\text{direct}} + \underbrace{\frac{1}{4}\sum_{n',n''} k^{(4)}_{\beta_n}(n; n', n'')C_{n'}C_{n''}}_{\text{quartic seed}} - \underbrace{\sum_{n'} \frac{k'_{nnnn'}}{2\omega_{n'}} r_{n'}}_{\text{quartic-}r\text{ block}} \\
& \underbrace{-4B\sum_{n',t}\cdots}_{\text{mixed prefactor block}} \quad \underbrace{+4\sum_{n',t}\left(X_{nt}X_{n't}F^{nn'} + X^{nt}X^{n't}F_{nn'}\right)}_{xf\text{ block}} \underbrace{-4\sum_{n',t}\left(X_{nt}X^{n't}F_{n'n} + X^{nt}X_{n't}F^{n'n}\right)}_{\text{cross block}} \\
& \underbrace{-16\sum_{n'}\left[U_{nn'}(\omega_n+\omega_{n'}) + V_{nn'}(\omega_{n'}-\omega_n)\right]}_{uv\text{ block}}.
\end{aligned}$$

This is exactly the block structure implemented in `/Users/vincenzobarone/centrifugal/linear_dv_aliev_terms.py`. The point of the present derivation is not to change the formula, but to make clear how the quartic and operator parts are represented in the code.

Interpretation as the linear H_{24} block. The semirigid-molecule derivation of Aliev and Watson (`/Users/vincenzobarone/Desktop/perottiche/Vibration-Rotation_Spectra_of_Semirigid_Molecules.pdf`, Section IV.B) states explicitly that H_{22} describes the vibrational dependence of the rotational constants, while H_{24} describes the vibrational dependence of the principal centrifugal-distortion constants. More importantly, the transformed fourth-order sector is not just the bare vibrational contact-transformation term. The explicit Table V formula for H_{24} already contains the linear-molecule ancestors of the present $X/F/U/V$ blocks. In the notation of Eq. (106) of that chapter, the final effective block then has the structure

$$\widetilde{H}_{24} = H_{24} + i[S_{03}, H_{12}] + i[S_{11}^{(R)}, H_{04}] + i[S_{13}^{(R)}, H_{02}],$$

up to the OCR ambiguities of the scanned text, but with the commutator structure itself unambiguous. Accordingly, the correct theoretical interpretation of the implemented β_n is not "the quartic term alone," but the linear-molecule realization of the effective $\widetilde{H}_{24}$ sector. In that interpretation:

- the quartic seed and quartic-r pieces are the H_{40}-driven anharmonic input to the effective constant;
- the mixed, xf, cross, and uv pieces belong already to the explicit Table V realization of the bare H_{24} kernel, before the subsequent block-diagonal corrections of Eq. (106) are applied;
- the practical problem of the present branch is therefore not whether β_n "should be quartic," but whether the current reduction of the *off-diagonal* operatorial part of the effective $\widetilde{H}_{24}$ sector is the correct one.

The consequence is important for the present note: the previous provisional reading of the parallel uv insertion as a direct proxy for

$$[S_{03}, H_{12}]$$

is too strong and should not be retained. Table V shows that the U/V sector already belongs to the bare H_{24} expression itself. The additional commutators of Eq. (106) are corrections *on top of* that explicit H_{24} kernel. The unresolved parallel residual is therefore not evidence that the code isolates a missing $[S_{03}, H_{12}]$ correction; it is evidence that the current linear reduction of the off-diagonal $U_{nn'}/V_{nn'}$ sector inside the explicit H_{24} kernel is still not fully understood.

This point is not just inferential. In the scanned Table V formula, the fourth-order block H_{24} contains in the same $(q_k q_l + p_k p_l)/2$ kernel:

- the R-type pieces,
- the X/F bilinears,
- and explicitly the term

$$-8 \sum_m \left\{ U_{km} U_{lm} (2\omega_m + \omega_k + \omega_l) + V_{km} V_{lm} (2\omega_m - \omega_k - \omega_l) \right\}.$$

So the present linear uv block is not an external add-on to H_{24}; it is already part of the bare H_{24} kernel written in Table V. The remaining open problem is therefore narrower than previously thought: it is the *linear reduction* of this explicit off-diagonal U/V sector, not the absence of a further contact-transformation correction.

There is also an important index distinction in Table V. The observable principal distortion constants come from the *diagonal* part of the external vibrational factor

$$\frac{1}{2}(q_k q_l + p_k p_l),$$

namely from the projection $k = l = n$. By contrast, higher-order l-doubling and rotational-resonance terms are carried by the off-diagonal external sectors $k \neq l$ (or by k, l being components of a degenerate pair). This means that the mere presence of an internal sum over mode labels in the U/V kernel does *not* by itself imply that the coefficient is a resonance term rather than a principal D-type constant. What matters is whether the *external* projection is diagonal.

For the present line of work this is crucial. The parallel coefficient β_n is supposed to represent the diagonal projection $k = l = n$ of the linear-molecule specialization of H_{24}. The unresolved issue is therefore not whether internal couplings to other modes should appear at all, but how the bare Table V structure

$$U_{km}U_{lm}, \qquad V_{km}V_{lm}$$

reduces, in the linear case, to the effective two-index objects used in the 1986 formulas and then in the present code. That precise reduction is still the missing derivation.

At this stage one point is already clear from the formulas themselves. The objects called $U_{nn'}$ and $V_{nn'}$ in the linear formulas of 1986 are *not* the same objects as the U_{km} and V_{km} that appear in Table V of the general semirigid-molecule derivation. In Table V the relevant fourth-order contribution enters as a product

$$U_{km}U_{lm}, \qquad V_{km}V_{lm},$$

inside the bare H_{24} kernel. By contrast, the linear formulas already use compact two-index descendants

$$U_{nn'}, \qquad V_{nn'},$$

which are themselves quadratic contractions over perpendicular modes,

$$U_{nn'}, V_{nn'} \;\propto\; \sum_t \zeta_{nt}\zeta_{n't}(\cdots).$$

So the 1986 notation is already a *post-contraction* notation. The real missing step is therefore the derivation of the compact linear descendants $U_{nn'}$ and $V_{nn'}$ from the product structure $U_{km}U_{lm}$ and $V_{km}V_{lm}$ of the general H_{24} expression.

The general chapter also states explicitly that one should work first with the nonlinear formulas and then deduce the linear ones, because this deduction is "comparatively easy," while for linear molecules only the xy block of the rotational tensors is retained. Taken together with the Table V formula, this shows where the linear reduction lives: not in a new perturbative term, but in the specialization of the general xy-block product

$$U_{km}U_{lm}, \qquad V_{km}V_{lm}$$

to the parallel/perpendicular split of a linear molecule. In other words, the compact 1986 objects $U_{nn'}$ and $V_{nn'}$ must be the result of a further linear-molecule contraction of the general xy-block kernel, not primitive objects at the same level of generality as the Table V quantities.

Explicit linear contraction of the xy block. In a linear molecule the external diagonal projection relevant to β_n is

$$k = l = n,$$

with n a parallel Σ mode. The internal mode label m that couples to this external Σ sector through the retained xy block belongs to a perpendicular Π pair. Writing the two Cartesian components of that pair in the physical Aliev basis as (t_a, t_b), the general xy-block product specializes as

$$U_{km}U_{lm} \;\longrightarrow\; U_{n,t_a}U_{n',t_a} + U_{n,t_b}U_{n',t_b},$$

and likewise

$$V_{km}V_{lm} \;\longrightarrow\; V_{n,t_a}V_{n',t_a} + V_{n,t_b}V_{n',t_b}.$$

Because the two components (t_a, t_b) span one physical perpendicular pair, this converts the general pair-component product into a single sum over perpendicular pairs. The Table IV definitions give

$$X_{kl} = 2(N_{kl} + M_{kl}), \qquad X^{kl} = 2(N_{kl} - M_{kl}),$$

so that

$$M_{kl} = \frac{X_{kl} - X^{kl}}{4}, \qquad N_{kl} = \frac{X_{kl} + X^{kl}}{4}.$$

After specializing to the retained xy block, each component of a physical Π_t pair contributes the same scalar frequency kernel and differs only by the sign convention carried by the Cartesian pair basis. The componentwise products therefore collapse to the pair invariants

$$U_{n,t_a}U_{n',t_a} + U_{n,t_b}U_{n',t_b} \;\longrightarrow\; \zeta_{nt}\zeta_{n't}\,\mathcal{U}(\omega_n, \omega_{n'}, \omega_t),$$

$$V_{n,t_a}V_{n',t_a} + V_{n,t_b}V_{n',t_b} \;\longrightarrow\; \zeta_{nt}\zeta_{n't}\,\mathcal{V}(\omega_n, \omega_{n'}, \omega_t).$$

Carrying the Table IV denominators $4(\omega_n + \omega_{n'})$ and $4(\omega_n - \omega_{n'})$ together with the $1/4$ factors from the substitutions for M_{kl} and N_{kl} yields exactly the overall coefficient $B^2/8$ of the compact linear descendants. In the notation of the present implementation, the contracted parallel objects are therefore

$$U_{nn'} = \frac{r_{nn'}}{4(\omega_n + \omega_{n'})} + \frac{B^2}{8}\sum_t \frac{\zeta_{nt}\zeta_{n't}}{\omega_t\sqrt{\omega_n\omega_{n'}}}\,\frac{(\omega_n + \omega_t)^2(\omega_{n'} - \omega_t)^2 + (\omega_n - \omega_t)^2(\omega_{n'} + \omega_t)^2}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)},$$

and, for $n \neq n'$,

$$V_{nn'} = \frac{r_{nn'}}{4(\omega_n - \omega_{n'})} + \frac{B^2}{8}\sum_t \frac{\zeta_{nt}\zeta_{n't}}{\omega_t\sqrt{\omega_n\omega_{n'}}}\,\frac{(\omega_n + \omega_t)^2(\omega_{n'} + \omega_t)^2(\omega_n + \omega_{n'} - 2\omega_t) - (\omega_n - \omega_t)^2(\omega_{n'} - \omega_t)^2(\omega_n + \omega_{n'} +}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)(\omega_n - \omega_{n'})}$$

with $V_{nn} = 0$ by antisymmetry. This is the compact two-index form used in the 1986 linear formulas and in the current code path. The essential point is that the compact descendants do not come from a mere renaming of the general semirigid-molecule symbols: they are the result of the explicit contraction of the retained xy block over the two components of each physical Π pair.

What Table IV already fixes exactly. The Table IV screenshots also make the intermediate hierarchy explicit:

$$M_{kl} = M_{lk} = \frac{R_k^l + R_l^k}{8(\omega_k + \omega_l)}, \qquad N_{kl} = N_{lk} = \begin{cases} \dfrac{R_k^l - R_l^k}{8(\omega_k - \omega_l)}, & \omega_k \neq \omega_l, \\ 0, & \omega_k \approx \omega_l, \end{cases}$$

and then

$$X_{kl} = 2(N_{kl} + M_{kl}), \qquad X^{kl} = 2(N_{kl} - M_{kl}).$$

So the general semirigid-molecule derivation already fixes the order of construction

$$R \;\longrightarrow\; (M, N) \;\longrightarrow\; X \;\longrightarrow\; (U, V).$$

The same Table IV screenshots also fix the general two-index families U_{kl} and V_{kl} themselves. In schematic notation,

$$U_{kl} = U_{lk} = \frac{R_{kl}^{\dagger} + \sum_m(M\text{-driven terms} - N\text{-driven terms}) + 4i[M_{kl}, H_{02}]}{4(\omega_k + \omega_l)},$$

while, for $\omega_k \neq \omega_l$,

$$V_{kl} = -V_{lk} = \frac{R^\dagger_{kl} + \sum_m (N\text{-driven terms} - M\text{-driven terms}) + 4i[N_{kl}, H_{02}]}{4(\omega_k - \omega_l)},$$

with $V_{kl} = 0$ in the near-degenerate limit. So the general theory does not leave the U/V sector undefined: it fixes it as the last stage of the same $R \to M, N \to X \to U, V$ ladder.

This is important for the linear case because it shows that the compact Aliev objects X_{nt} and X^{nt} are not independent phenomenological insertions. They are the linear descendants of the same Table IV ladder, specialized to the retained xy block and then rewritten in the parallel/perpendicular notation of the 1986 paper.

Accordingly, the status of the X block is no longer open, and neither is the structural origin of the compact $U_{nn'}$ and $V_{nn'}$ objects. What remains open is not the contraction itself. The remaining issue is only observational: whether the off-diagonal branch produced by the compact linear formulas is already the fully reduced observable contribution to β_n, or whether the present working implementation still overweights that branch numerically.

Quartic branch: general formula first. For the parallel coefficient, the quartic sector enters only through two insertions:

1. the quartic seed term
$$\frac{1}{4} \sum_{n',n''} k^{(4)}_{\beta_n}(n; n', n'')\, C_{n'} C_{n''},$$
2. the quartic term coupled to $r_{n'}$,
$$-\sum_{n'} \frac{k'_{nnnn'}}{2\omega_{n'}}\, r_{n'}.$$

At the level of the printed Aliev formula, the natural object is the full four-index quartic tensor. The code therefore treats the full representation as the primary one, and only afterwards defines the reduced practical backend.

Full branch. If the complete four-index quartic tensor is available, the parallel quartic terms are read directly from

$$k'_{ijkl}.$$

In this branch the seed term is implemented as

$$\frac{1}{4} \sum_{n',n''} k'_{nnn'n''}\, C_{n'} C_{n''},$$

and the quartic-r term as

$$-\sum_{n'} \frac{k'_{nnn'n'}}{2\omega_{n'}}\, r_{n'}.$$

So the full branch is simply the direct implementation of the four-index Aliev quartic objects in the parallel sector.

Reduced branch. If the full four-index tensor is not available, the code accepts instead the three-index reduced object

$$\texttt{k4\_reduced}(i, j, k) \equiv k'_{ii\,jk},$$

with the trailing pair canonically sorted. In this branch the same parallel formula is kept, but every quartic access is routed through the map

$$k'_{ijkl} \longmapsto \texttt{k4\_reduced}$$

whenever the requested index pattern is representable in the reduced form. In other words:

- the *formula* for β_n is not changed;
- only the storage model for quartic constants changes;
- `full` is the general branch;
- `reduced` is the practical branch when the four-index tensor is too expensive or unavailable.

This is the correct interpretation of the current implementation. The reduced branch is not a different Aliev formula. It is the same parallel quartic sector written through a cheaper storage model.

Implementation recommendation. The equations should therefore be organized around a single quartic theory, namely the full four-index tensor k'_{ijkl}. The reduced three-index branch should be treated only as an input/storage adapter:

- if full quartic data are available, they populate the backend directly;
- if only reduced data are available, they populate the representable entries of the same four-index backend;
- all non-representable quartic entries are then set to zero explicitly in that reduced-input branch.

So the recommended design is not two separate quartic implementations, but one general four-index implementation together with a reduced-input adapter. In this sense, the present equations do support the full quartic field, while the benchmarks reported here were simply run with the reduced three-index input.

Layer	Object	Role
Theory	full k'_{ijkl} field	unique general quartic implementation
Input adapter	`k4_reduced`(i,j,k)	reduced three-index population of representable entries
Benchmark branch here	reduced-input run	practical data branch used in the present tests

Meaning of $k^{(4)}_{\beta_n}(n;n',n'')$. The notation

$$k^{(4)}_{\beta_n}(n;n',n'')$$

used in this note is therefore only a compact way to denote the quartic object that feeds the seed part of β_n. Concretely:

$$k^{(4)}_{\beta_n}(n;n',n'') = \begin{cases} k'_{nnn'n''}, & \text{full branch,} \\ \texttt{k4\_reduced}(n,n',n''), & \text{reduced branch.} \end{cases}$$

So the implemented parallel quartic seed is a single symbolic formula with two storage backends, not two different theoretical formulas.

Parallel operator reduction. The current working branch retains the parallel uv insertion:

$$(\beta_n)_{\text{working}} \text{ keeps } -16\sum_{n'}\left(U_{nn'}(\omega_n+\omega_{n'})+V_{nn'}(\omega_{n'}-\omega_n)\right).$$

The operator part of β_n depends on the scalar objects entering X^{nt}, X_{nt}, $F^{nn'}$, $F_{nn'}$, $U_{nn'}$, and $V_{nn'}$. In the current linear branch these scalars are not read from the full degenerate Coriolis block through `principal_direction`, because that reduction returns the norm of the block and overdrives the parallel $U_{nn'}/V_{nn'}$ sector. They are instead built with

$$\texttt{zeta\_reduction} = \texttt{pair\_offdiag},$$

that is, from the off-diagonal symmetric component of the canonicalized pair block. The reason is structural: β_n needs component-type operator couplings, not the invariant norm of the degenerate block. This is why the parallel operator branch is now numerically sane only with `pair_offdiag`. In this sense, the present derivation of β_n contains two independent implementation choices:

- a quartic storage choice: `full` or `reduced`;
- an operator reduction choice: `pair_offdiag`, not `principal_direction`.

6. Perpendicular coefficient β_t

The implemented perpendicular coefficient is

$$\begin{aligned}
\beta_t = {} & \frac{1}{4}\sum_{n,n'} k^{(4)}_{\beta_t}(t;n,n')C_nC_{n'} + \mathcal{P}_t - \sum_n \frac{k^{(3)}_{ttn}}{2\omega_n} r_n \\
& - 4B\sum_{n,n'} C_n\zeta_{nt}\frac{2\omega_t^2\omega_{n'}^{3/2}C_{n'}\zeta_{n't} + (3\omega_t^2+\omega_{n'}^2)\omega_{n'}^{3/2}C_{n'}r_{nn'}}{\omega_t\sqrt{\omega_{n'}}(\omega_t^2-\omega_{n'}^2)} \\
& + 4\sum_{n,n'}\left(X_{nt}X_{n't}F^{nn'} + X^{nt}X^{n't}F_{nn'}\right) - 4\sum_{n,t'}\left(X_{nt}X^{nt'}F_{tt'} + X^{nt}X_{nt'}F^{tt'}\right) \\
& - 16\sum_{t'}\left[U_{tt'}(\omega_t+\omega_{t'}) + V_{tt'}(\omega_{t'}-\omega_t)\right].
\end{aligned} \tag{14}$$

Here the term multiplying r_n is *cubic*, not quartic. In the code this object is the pair-resolved cubic input carried by `k3_perp_pair[t][t][n]`; the notation above is therefore

$$k^{(3)}_{ttn},$$

not a four-index quartic constant. The previous notation with k'_{ttn} was a bad shorthand and should be regarded as a corrected typo in this note.

The formula written above for $V_{tt'}$ is obtained in exactly the same formal way used in the previous paragraph for $U_{tt'}$: start from the printed $V_{nn'}$ kernel, apply

$$n,n' \mapsto t,t', \qquad \sum_t \mapsto \sum_n,$$

and leave the internal frequency pattern otherwise unchanged. In this note $V_{tt'}$ is therefore not presented as an explicitly printed Aliev formula, but as the controlled formal-substitution branch generated from $V_{nn'}$. The code path in `/Users/vincenzobarone/centrifugal/linear_dv_aliev_terms.py` has been aligned to that exact substitution rule.

Bending seed currently used. The currently frozen diagnostic branch uses

$$\mathcal{P}_t = \texttt{rotder\_seed\_gram},$$

namely the seed built from the geometry/mode-based rotational-derivative scaffold in `/Users/vincenzobarone/centrifugal/linear_aliev_rotder_zeta.py`.

Final working sign convention. After correcting the negative X/F block to use the perpendicular families $F_{tt'}$ and $F^{tt'}$ rather than reusing $F_{nn'}$ and $F^{nn'}$, the remaining bending discrepancy is still controlled by the common sign of the full perpendicular $xf/cross$ insertion. In the present linear branch this is encoded as

$$\texttt{beta\_t\_xf\_cross\_sign} = -1.$$

This sign is not claimed here as a completed derivation from the printed Aliev formula. It is the only sign choice found so far that makes the linear-branch results physically reasonable after the remaining structural alternatives inside the implemented branch have been tested and eliminated. In particular, the code path explored and ruled out:

- the incorrect reuse of $F_{nn'}/F^{nn'}$ in the negative perpendicular block;
- the corrected $F_{tt'}/F^{tt'}$ structure with the printed overall sign;
- swaps of upper/lower X and F associations inside the same block.

None of those alternatives produced a physically satisfactory branch. The working sign

$$\texttt{beta\_t\_xf\_cross\_sign} = -1$$

is therefore retained explicitly as the only branch that restored physically reasonable bending behavior in the present implementation.

7. Optical scalar L

The implemented quartic optical scalar is

$$L = -\frac{8D_J^3}{B^2} + \frac{1}{24}\sum_{n,n',n'',n'''} k'_{nn'n''n'''}C_nC_{n'}C_{n''}C_{n'''} + 8BD_J\sum_n \frac{C_n^2}{\omega_n} - \sum_n \frac{r_n^2}{2\omega_n}. \tag{15}$$

This is exposed programmatically by

$$\texttt{build\_explicit\_aliev\_L\_model(inputs).value}.$$

The implementation also provides a bookkeeping partition of L into modewise contributions, but this partition remains provisional because the global $-8D_J^3/B^2$ term is not intrinsically mode-local.

Numerical check on HCN. In the current linear branch, HCN gives

$$B = 44193.2932754 \text{ MHz}, \qquad D = 0.077504109 \text{ MHz}, \qquad H = 6.526711222 \times 10^{-8} \text{ MHz},$$

$$L = -1.9070056626 \times 10^{-6} \text{ Hz}, \qquad D_v(0) = 0.0781112547 \text{ MHz}.$$

The current branch yields

$$\beta_{\parallel} \approx (-7.4221466811 \times 10^{-4},\ -4.7207680957 \times 10^{-4}) \text{ MHz},$$

$$\beta_\perp \approx (5.9366573168 \times 10^{-14})\ \text{MHz}.$$

Converting L to MHz and forming the dimensionless ratio,

$$\frac{L\,B^2}{D^3},$$

one obtains numerically

$$\frac{L\,B^2}{D^3} \approx -8.0000100305.$$

So, for HCN, the implemented optical constant satisfies the expected dominant scaling

$$L \approx -8\,\frac{D^3}{B^2}$$

to numerical precision. This is one of the cleanest internal consistency checks currently available for the linear branch.

Numerical check on C_2H_2. In the same working branch, C_2H_2 gives

$$B = 35274.2053898\ \text{MHz}, \qquad D = 0.043438118\ \text{MHz}, \qquad H = 3.305071123 \times 10^{-8}\ \text{MHz},$$

$$L = -5.2697369299 \times 10^{-7}\ \text{Hz}, \qquad D_v(0) = 0.0441037909\ \text{MHz}.$$

The current branch yields

$$\beta_\parallel \approx (-6.9857171750 \times 10^{-4},\ -1.3950714528 \times 10^{-4},\ -4.7283759116 \times 10^{-4})\ \text{MHz},$$

$$\beta_\perp \approx (7.9642614003 \times 10^{-14},\ -1.0214632548 \times 10^{-5})\ \text{MHz}.$$

Unlike HCN and OCS, the main role of C_2H_2 in the present audit is not the optical scaling alone, but the fact that it exposes the residual off-diagonal parallel channel structure with enough numerical resolution to isolate the dominant

$$\Sigma_g^+ \longleftrightarrow \Sigma_g^+$$

operator coupling.

Numerical check on HCCD. In the same working branch, HCCD gives

$$B = 29682.4448229\ \text{MHz}, \qquad D = 0.030190776\ \text{MHz}, \qquad H = 2.175593934 \times 10^{-8}\ \text{MHz},$$

$$L = -2.4987010319 \times 10^{-7}\ \text{Hz}, \qquad D_v(0) = 0.0305966553\ \text{MHz}.$$

The current branch yields

$$\beta_\parallel \approx (-1.3534963171 \times 10^{-4},\ -3.7968269603 \times 10^{-4},\ -2.9262219593 \times 10^{-4})\ \text{MHz},$$

$$\beta_\perp \approx (7.2130392340 \times 10^{-14},\ -2.0520545655 \times 10^{-6})\ \text{MHz}.$$

The corresponding optical scaling check is again consistent with the dominant Aliev form:

$$\frac{L\,B^2}{D^3} \approx -8.0000150994.$$

Numerical check on OCS. In the same working branch, OCS gives

$$B = 6001.3622546 \text{ MHz}, \qquad D = 0.001227766 \text{ MHz}, \qquad H = -1.020490762 \times 10^{-10} \text{ MHz},$$
$$L = -4.1109022006 \times 10^{-10} \text{ Hz}, \qquad D_v(0) = 0.0834577459 \text{ MHz}.$$

The current branch yields

$$\beta_{\|} \approx (-8.2230860790 \times 10^{-2},\ -8.2229096822 \times 10^{-2}) \text{ MHz},$$
$$\beta_{\perp} \approx (-1.1147103672 \times 10^{-9}) \text{ MHz}.$$

Again, the optical scaling check is essentially exact:

$$\frac{L\,B^2}{D^3} \approx -8.00000065269.$$

So the optical branch remains internally consistent also in OCS, even though the parallel vibrational increments themselves are not yet closed.

Benchmark summary table. The four benchmark species can be summarized by the compactness diagnostic

$$B_{\max}^{\text{mode}} = \max_n |\beta_n^{\text{mode}}|, \qquad B_{\max}^{\text{pair}} = \max_{n<n'} |\Gamma_{nn'}|, \qquad \rho = \frac{B_{\max}^{\text{pair}}}{B_{\max}^{\text{mode}}}.$$

The scalar compact branch is admissible only when $\rho \ll 1$.

Species	LB^2/D^3	$B_{\max}^{\text{mode}}$ (cm^{-1})	ρ	Compact branch
HCN	-8.0000100305	4.6305×10^{-12}	0	admissible
HCCD	-8.0000150994	1.5647×10^{-12}	0	admissible
C_2H_2	-8.0000078253	2.3069×10^{-12}	4.8767×10^{3}	not admissible
OCS	-8.00000065269	6.4020×10^{-14}	4.2844×10^{7}	not admissible

This table condenses the practical conclusion of the whole audit. The optical branch is uniformly stable, while the compact scalar parallel branch is acceptable only in the species where the pair sector is numerically negligible.

8. Final state dependence

The state dependence implemented for linear molecules is

$$D_v = D_J - \sum_n \beta_n \left(v_n + \frac{1}{2}\right) - \sum_t \beta_t (v_t + 1). \tag{16}$$

9. Working linear branch

The current working branch for linear molecules is:

- bending seed:
  $$\texttt{pair\_seed\_source} = \texttt{rotder\_seed\_gram};$$
- operator Coriolis reduction:
  $$\texttt{zeta\_reduction} = \texttt{pair\_offdiag};$$
- perpendicular *xf/cross* sign:
  $$\texttt{beta\_t\_xf\_cross\_sign} = -1;$$

With this branch the two current linear benchmarks in the repo behave as follows:

C_2H_2.

$$v_1 \sim -1.125 \times 10^{-8}, \qquad v_2 \sim +1.934 \times 10^{-13}, \qquad v_3 \sim -1.125 \times 10^{-8},$$
$$v_4 \sim +3.463 \times 10^{-19}, \qquad v_5 \sim +9.584 \times 10^{-13}.$$

HCN.

$$\beta_{\parallel} \sim 10^{-18} \text{ to } 10^{-12}, \qquad \beta_{\perp} \sim 10^{-22}.$$

The important point is not that these are yet a final literature match. The important point is that the catastrophic scale failures of the earlier branch are gone.

10. Why `pair_offdiag` In The Parallel U/V Sector

For each parallel mode n and degenerate perpendicular pair t, the Gaussian-side Coriolis information entering the operator sector can be written as a real 2×2 block

$$M_{nt} = \begin{pmatrix} \zeta_{x,n,t_a} & \zeta_{x,n,t_b} \\ \zeta_{y,n,t_a} & \zeta_{y,n,t_b} \end{pmatrix}.$$

In the linear benchmarks used here, these blocks are numerically almost traceless and almost symmetric, so to excellent approximation one can write

$$M_{nt} \approx \begin{pmatrix} a_{nt} & s_{nt} \\ s_{nt} & -a_{nt} \end{pmatrix}.$$

Under a rotation of the degenerate pair basis by an angle θ,

$$M'_{nt} = R(\theta)^T M_{nt} R(\theta),$$

the coefficients transform as

$$a'_{nt} = a_{nt} \cos 2\theta + s_{nt} \sin 2\theta, \qquad s'_{nt} = -a_{nt} \sin 2\theta + s_{nt} \cos 2\theta.$$

So the pair (a_{nt}, s_{nt}) is a spin-2 doublet under rotations of the degenerate subspace, while the invariant norm

$$\rho_{nt} = \sqrt{a_{nt}^2 + s_{nt}^2}$$

is not a component but the norm of the whole block.

This is why the older reduction `principal_direction` is not admissible for the operator families $X/F/U/V$: it effectively feeds the scalar operator sector with ρ_{nt} rather than with a component-type quantity ζ_{nt}. The scanned Aliev notation is component-like ($\zeta^x_{n,tb}$-type), not norm-like.

Numerically the consequence is decisive:

- with `principal_direction`, the off-diagonal $U_{nn'}/V_{nn'}$ block drives `uv_parallel` to about 10^{-3} on C_2H_2 and to about 10^{-7} on HCN;
- with `pair_offdiag`, the same block drops to the physically sane $10^{-8} \dots 10^{-12}$ range.

Lemma. Let

$$M = \begin{pmatrix} a & s \\ s & -a \end{pmatrix}$$

be the symmetric traceless operator block associated with a degenerate pair in its canonicalized two-dimensional subspace. Under a rotation

$$M \mapsto R(\theta)^T M R(\theta),$$

the coefficients (a, s) transform as a two-component object rotated by the double angle 2θ, while the norm $\rho = \sqrt{a^2 + s^2}$ is invariant. It follows that:

1. `principal_direction` returns $\pm\rho$, hence the norm of the block and not one of its components;
2. `pair_offdiag` returns the component s in the chosen canonical basis;
3. only a component-type extraction is compatible with the Aliev notation $\zeta^x_{n,tb}$, which is component-valued rather than norm-valued.

Therefore the current working branch keeps the parallel $U_{nn'}/V_{nn'}$ sector active, but feeds the operator families with the off-diagonal canonical component

$$\texttt{pair\_offdiag} = \frac{M_{12} + M_{21}}{2},$$

rather than with the block norm selected by `principal_direction`.

This does prove, within the current implementation framework, that `principal_direction` is the wrong object for the operator sector and that `pair_offdiag` is the correct working replacement in the canonical pair basis used throughout the present linear reduction.

Analytic consequence for the off-diagonal parallel sector. The previous lemma also shows why the remaining parallel mismatch cannot be settled by a scalar renormalization of `pair_offdiag`. For each parallel mode n and perpendicular pair t, the retained xy block carries not one scalar but a spin-2 doublet

$$\mathbf{z}_{nt} = (a_{nt}, s_{nt}),$$

where a_{nt} is the traceless diagonal component and s_{nt} is the symmetric off-diagonal component of the canonical pair block. Under a common rotation of the pair basis this doublet transforms covariantly, so any observable reduction for a pair of parallel modes (n, n') must be built from the bilinear invariants of two such doublets. There are only two natural bilinears:

$$\mathbf{z}_{nt} \cdot \mathbf{z}_{n't} = a_{nt}a_{n't} + s_{nt}s_{n't},$$

which is symmetric under $n \leftrightarrow n'$, and

$$\mathbf{z}_{nt} \wedge \mathbf{z}_{n't} = a_{nt}s_{n't} - s_{nt}a_{n't},$$

which is antisymmetric under $n \leftrightarrow n'$.

This matches exactly the general symmetry pattern of the compact linear objects:

$$U_{nn'} = U_{n'n}, \qquad V_{nn'} = -V_{n'n}.$$

So the off-diagonal parallel branch cannot, in the fully reduced theory, depend on one and the same scalar insertion $\zeta_{nt}\zeta_{n't}$ for both families. The symmetric $U_{nn'}$ channel must reduce through the dot-type invariant, while the antisymmetric $V_{nn'}$ channel must reduce through the wedge-type invariant. The present working implementation instead feeds both channels with the same scalar product

$$\zeta_{nt}\zeta_{n't},$$

where ζ_{nt} is identified with the single component extracted by `pair_offdiag`. That identification can be serviceable on the diagonal $n = n'$ sector, where the antisymmetric invariant vanishes identically and the observable branch depends only on one self-contraction, but it is not analytically sufficient for the off-diagonal sector $n \neq n'$.

This is the sharpest formulation of the residual problem: the compact 1986 notation writes the off-diagonal branch with scalar descendants $U_{nn'}, V_{nn'}$, but the general-to-linear reduction shows that these descendants must come from two distinct bilinear contractions of the spin-2 pair doublets. The present use of a single scalar `pair_offdiag` therefore under-resolves the analytic content of the off-diagonal $U_{nn'}/V_{nn'}$ sector.

An immediate consequence is that the most naive "completion" of the working branch is already excluded. If one feeds the symmetric channel $U_{nn'}$ with the full spin-2 dot product

$$a_{nt}a_{n't} + s_{nt}s_{n't},$$

while feeding the antisymmetric channel $V_{nn'}$ with the wedge product above, the parallel branch of C_2H_2 returns at once to the old unphysical 10^{-3} scale. So the observable off-diagonal branch cannot be the raw symmetric bilinear of the canonical pair doublets either. The remaining analytic problem is therefore more specific: the compact linear descendant must retain a component-level reduction in the symmetric channel as well, but not the same undifferentiated scalar currently used for both $U_{nn'}$ and $V_{nn'}$.

This sharpens the overall status of the derivation. The general-to-linear reduction itself is no longer the weak point: it already fixes the existence of two distinct off-diagonal bilinear channels, one symmetric and one antisymmetric. What remains unresolved is therefore internal to the compact 1986 branch. Either the printed compact formulas suppress an additional component-level projection in the symmetric channel, or the present working identification of the compact scalar descendant with the single extracted component `pair_offdiag` is still too crude for $n \neq n'$.

The benchmark set shows that this is not a minor tuning issue but a structural one. Writing a canonical pair block as

$$M = d\,I + k\,J + a\,K + s\,L,$$

with I the identity, J the skew generator, and (K, L) the traceless symmetric spin-2 sector, one finds:

- in HCN and HCCD the active blocks are dominated by the (d, k) sector;
- in C_2H_2 and OCS the active blocks are dominated by the (a, s) sector.

The scalar reductions implemented in the code probe these sectors differently:

- `pair_diag` extracts the scalar part d;
- `pair_offdiag` extracts the symmetric off-diagonal part s;
- `principal_direction` returns the norm of the spin-2 block.

The four benchmarks then exclude every universal scalar choice:

- `pair_diag` suppresses the problematic channels in C_2H_2/OCS but blows up HCCD;
- `pair_offdiag` keeps HCN/HCCD under control but leaves a systematic residual in C_2H_2/OCS;
- `principal_direction` is wrong everywhere for the operator sector.

Therefore the last unresolved point is now sharply identified: the off-diagonal parallel branch of the compact 1986 formulas cannot be represented by a single universal scalar descendant of the 2×2 pair block. The compact notation necessarily hides at least one additional component-level distinction between the scalar/skew sector and the traceless-symmetric sector of the canonical pair block.

The same conclusion survives the next obvious refinement. One may try to feed the symmetric channel $U_{nn'}$ and the antisymmetric channel $V_{nn'}$ with different scalar descendants of the pair block, for example one component from the (d, k) sector and another from the (a, s) sector. This also fails. In the present benchmark set all natural mixed assignments of this kind either reproduce the old blow-up on HCN/HCCD or destroy the partial cancellation that keeps C_2H_2/OCS below the 10^{-3} regime. So the unresolved branch is not repaired by assigning one universal scalar to U and a different universal scalar to V.

This leaves a single consistent reading of the residual problem. The compact 1986 off-diagonal notation is still suppressing more structure than a single pair scalar can carry. In other words, the unresolved branch must be lifted from a scalar reduction of the pair block to a genuinely component-resolved off-diagonal object before a fully general implementation can be claimed.

At the same time, the new decompacted weighted audit shows something stronger than the earlier purely structural argument. In HCN the scalar/skew sector (d, k) would generate a sizeable off-diagonal U contribution if it were retained, yet the working compact branch and the observed smallness of the parallel correction both suppress it. In OCS and C_2H_2, by contrast, the full spin-2 dot product $aa' + ss'$ is much larger than the actual compact branch. So the compact linear notation is not using the scalar/skew sector, and it is not using the full traceless-symmetric bilinear either. It is selecting precisely the symmetric off-diagonal component s of the canonical pair block, that is, the same component carried operationally by `pair_offdiag`.

This closes one part of the ambiguity. The compact 1986 off-diagonal branch is not an arbitrary scalarization of the pair block: it is an s-only projection. What remains open after this identification is no longer the meaning of the compact scalar itself, but only the quantitative fact that the resulting s-projected branch is still too large on the C_2H_2/OCS benchmarks.

The last obvious scalar rescue also fails. Writing the observable off-diagonal branch schematically as

$$\Delta\beta_n^{(uv)} = \lambda_U \, U\text{-part} + \lambda_V \, V\text{-part},$$

one finds that no universal choice of λ_U or λ_V can suppress the residual across all benchmarks. The values required to cancel the dominant channel vary strongly from species to species: for instance the OCS channels would require $\lambda_V \approx 0.712$, while the dominant C_2H_2 channels require $\lambda_V \approx 1.53$, and HCN/HCCD do not admit any coherent common value. So the remaining mismatch is not a missed global prefactor in front of $U_{nn'}$ or $V_{nn'}$.

At that point the residual ambiguity is fully localized. The compact 1986 off-diagonal notation has the correct qualitative selector—the s component of the canonical pair block—but it is still not rich enough to produce a general quantitative observable branch. The only consistent next step is therefore an explicit component-resolved reformulation of the off-diagonal parallel sector, rather than another scalar modification of the compact formula.

That reformulation is now the correct implementation stance. In the codebase, the compact branch is retained as the literal 1986-style scalar realization, but the general linear observable is exposed in decompacted form: genuinely mode-anchored parallel terms stay in β_n, whereas the unresolved off-diagonal channels are carried explicitly as unordered pair contributions (n, n'). This does not by itself remove the quantitative mismatch on C_2H_2/OCS, but it does remove the false claim that the whole linear branch has been captured by one-mode coefficients alone. From an implementation point of view, this is the first general form that is both executable and honest about what remains compact and what does not.

Final derivation statement for the linear branch. At this point the derivation can be stated in a closed form. Starting from the general Table IV objects

$$R_{kl}, \qquad M_{kl}, \qquad N_{kl}, \qquad X_{kl}, \qquad X^{kl}, \qquad U_{kl}, \qquad V_{kl},$$

and from the bare H_{24} kernel of Table V, the linear case is obtained by:

1. restricting the rotational sector to the xy block;
2. separating external Σ modes from perpendicular Π pairs;
3. projecting the observable D_v branch on the external diagonal sector $k = l = n$;
4. contracting the internal Π_t doublet over its two Cartesian components (t_a, t_b).

After this reduction, the linear observable splits unavoidably into two different objects:

- a mode-anchored part, which is legitimately collected into one-mode coefficients β_n and β_t;
- a parallel off-diagonal part, generated by unordered parallel pairs (n, n'), which remains pair-anchored even after the linear xy specialization.

Therefore the general linear observable is not, in full generality, a function of one-mode coefficients alone. Its correct reduced form is

$$D_v = D_J - \sum_n \beta_n^{\text{mode}} f_n(v) - \sum_t \beta_t f_t(v) - \sum_{n<n'} \Gamma_{nn'} \left(f_n(v) + f_{n'}(v)\right),$$

where $f_n(v) = v_n + \frac{1}{2}$ for parallel modes and $f_t(v) = v_t + 1$ for perpendicular pairs. The compact 1986 formulas correspond to the further compression

$$\Gamma_{nn'} \mapsto \text{absorbed into one-mode descendants},$$

which is operationally acceptable only when the off-diagonal pair sector is numerically negligible. In the benchmark set studied here this is true for HCN/HCCD, but not for C_2H_2/OCS. This is the definitive reason why the decompacted observable, not the compact scalar branch, is the general final linear implementation.

Corollary: when the compact 1986 branch is legitimate. The previous statement yields an immediate practical criterion. Let

$$B_{\max}^{\text{mode}} = \max_n |\beta_n^{\text{mode}}|, \qquad B_{\max}^{\text{pair}} = \max_{n<n'} |\Gamma_{nn'}|.$$

Then the compact 1986 branch is observationally legitimate only in the regime

$$B_{\max}^{\text{pair}} \ll B_{\max}^{\text{mode}},$$

because only in that limit can the pair-anchored parallel sector be absorbed into effective one-mode descendants without changing the dominant structure of D_v. This is exactly the situation found for HCN and HCCD. For C_2H_2 and OCS one instead finds

$$B_{\max}^{\text{pair}} \gtrsim B_{\max}^{\text{mode}},$$

so the compact branch is not a faithful final observable representation even though it remains a useful legacy shorthand.

Residual parallel channel in C_2H_2. With the current working branch, the remaining parallel uv contribution in C_2H_2 is not a diffuse background over all mode pairs. It is dominated almost entirely by the single off-diagonal channel

$$n = 0 \longleftrightarrow n' = 2.$$

These two parallel modes carry the same symmetry label,

$$\Sigma_g^+ \longleftrightarrow \Sigma_g^+,$$

and the corresponding operator contribution is driven almost entirely by the first perpendicular pair, whose symmetry is

$$\Pi_g,$$

while the second perpendicular pair

$$\Pi_u$$

is numerically inactive in the same block.

So the remaining parallel uv term is not a symmetry-forbidden artifact and not a spurious consequence of the `pair_offdiag` reduction. It is a real, symmetry-allowed operator channel already present in the raw pair block. The unresolved issue is therefore narrower: whether the magnitude with which the B^2 operator piece enters U_{02}/V_{02} is exactly the one required by the original Aliev derivation.

Exact cancellation of the $r_{nn'}$ piece for $n \neq n'$. For off-diagonal parallel pairs, the final uv insertion is

$$-16\left[U_{nn'}(\omega_n + \omega_{n'}) + V_{nn'}(\omega_{n'} - \omega_n)\right].$$

Substituting the definitions of $U_{nn'}$ and $V_{nn'}$ shows that the entire $r_{nn'}$ contribution cancels identically:

$$-16\left[\frac{r_{nn'}}{4(\omega_n + \omega_{n'})}(\omega_n + \omega_{n'}) + \frac{r_{nn'}}{4(\omega_n - \omega_{n'})}(\omega_{n'} - \omega_n)\right] = 0.$$

Therefore, for $n \neq n'$, the surviving parallel uv block is a pure $B^2\zeta\zeta$ operator term.

In particular, for one perpendicular pair t, the surviving off-diagonal contribution can be written as

$$(\beta_n^{uv})_{nn';t} = -2B^2\,\frac{\zeta_{nt}\zeta_{n't}}{\omega_t\sqrt{\omega_n\omega_{n'}}}\left[(\omega_n + \omega_{n'})\,\mathcal{U}_{nn';t} + (\omega_{n'} - \omega_n)\,\mathcal{V}_{nn';t}\right],$$

with

$$\mathcal{U}_{nn';t} = \frac{(\omega_n + \omega_t)^2(\omega_{n'} - \omega_t)^2 + (\omega_n - \omega_t)^2(\omega_{n'} + \omega_t)^2}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)},$$

$$\mathcal{V}_{nn';t} = \frac{(\omega_n + \omega_t)^2(\omega_{n'} + \omega_t)^2(\omega_n + \omega_{n'} - 2\omega_t) - (\omega_n - \omega_t)^2(\omega_{n'} - \omega_t)^2(\omega_n + \omega_{n'} + 2\omega_t)}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)(\omega_n - \omega_{n'})}.$$

This matters because it shows that the residual parallel problem is no longer a quartic/cubic contamination issue. In the current working branch it is a pure off-diagonal operator issue inside the $B^2\zeta\zeta$ sector.

Closed off-diagonal kernel and what OCS changes. For $n \neq n'$, after the cancellation of the $r_{nn'}$ contribution, the single-channel contribution of one perpendicular pair t to the final parallel-sector uv block can be written in closed form as

$$\Delta\beta_n^{(uv)}(n, n'; t) = -4B^2\,\zeta_{nt}\zeta_{n't}\,K(x, y),$$

with

$$x = \frac{\omega_n}{\omega_t}, \qquad y = \frac{\omega_{n'}}{\omega_t},$$

and

$$K(x,y)=\frac{x^3y^2-2x^3y+x^3+x^2y^3-2x^2y^2-3x^2y-2xy^3-3xy^2+4xy+x+y^3+y+2}{\sqrt{xy}(x-1)(x+1)(y-1)(y+1)}.$$

Thus the residual off-diagonal channel is not a generic quartic artifact; it is a fully explicit frequency-ratio kernel multiplying the operator product $B^2\zeta_{nt}\zeta_{n't}$.

Moreover, this kernel is exactly symmetric:

$$K(x,y)=K(y,x).$$

Hence the off-diagonal contribution generated by the unordered pair $\{n,n'\}$ is the same in β_n and in $\beta_{n'}$. In other words, the current uv insertion treats the channel as a pair contribution distributed equally onto the two one-mode coefficients. This is a useful structural fact: the residual term is not mode-anchored, but pair-anchored.

In C_2H_2, for the dominant channel $n=0\leftrightarrow 2$ mediated by the first Π_g pair, one finds

$$x\approx 3.90,\qquad y\approx 6.62,\qquad K(x,y)\approx 2.953.$$

This is why the C_2H_2 residual looked numerically compatible with a modest renormalization of the operator component.

The OCS case changes the interpretation decisively. In OCS the dominant channel is $n=0\leftrightarrow 1$, mediated by the only Π pair, and one gets

$$x\approx 1.71,\qquad y\approx 4.18,\qquad K(x,y)\approx -4.925.$$

So one of the two parallel frequencies lies much closer to the threshold $\omega_n=\omega_t$, and the same off-diagonal $B^2\zeta\zeta$ mechanism is amplified by a more severe frequency-ratio kernel. This rules out the idea that the remaining parallel discrepancy can be repaired by a universal simple renormalization such as $s\mapsto s/\sqrt{2}$ or $s\mapsto s/2$.

This also clarifies what the problem is *not*. The symmetry $K(x,y)=K(y,x)$ suggests that one should at least examine whether the off-diagonal channel is being assigned to one-mode coefficients β_n in a way that double-counts an unordered pair contribution. However, a naive global factor 1/2 on the off-diagonal uv sector does not close the benchmarks: OCS remains far too large even after that reduction. So the remaining issue is not a mere combinatorial duplication of the unordered pair. It is the more substantive question of whether the full pair-anchored operator term $-4B^2\zeta_{nt}\zeta_{n't}K(x,y)$ belongs, without further projection or repartition, inside the final one-mode coefficients β_n.

In fact, because the implemented state dependence is linear,

$$D_v=D_J-\sum_n\beta_n\left(v_n+\frac{1}{2}\right)-\sum_t\beta_t(v_t+1),$$

any symmetric off-diagonal channel assigned equally to β_n and $\beta_{n'}$ is algebraically equivalent to a separate pair contribution

$$-\gamma_{nn'}\left[\left(v_n+\frac{1}{2}\right)+\left(v_{n'}+\frac{1}{2}\right)\right]=-\gamma_{nn'}(v_n+v_{n'}+1).$$

So the residual discrepancy cannot be removed merely by changing notation from "equal additions to β_n and $\beta_{n'}$" to "explicit pair term." The placement ambiguity is only interpretive. The quantitative problem remains the same: in the present implementation the pair-anchored operator amplitude itself is too large, especially in OCS.

At the current stage the unresolved point in β_n is therefore no longer a small normalization issue of `pair_offdiag`. It is the off-diagonal operator sector $U_{nn'}/V_{nn'}$ itself, more precisely the way the $B^2\zeta_{nt}\zeta_{n't}$ kernel enters the final parallel sum for $n\neq n'$.

What is left in the prefactor. The bookkeeping can also be written in a way that isolates the only remaining scalar freedom. If the off-diagonal pieces of $U_{nn'}$ and $V_{nn'}$ are written with a common prefactor a,

$$U_{nn'}^{(B^2)} = a\,B^2(\cdots), \qquad V_{nn'}^{(B^2)} = a\,B^2(\cdots),$$

then the final off-diagonal contribution to β_n takes the form

$$\Delta\beta_n^{(uv)}(n,n';t) = -32a\,B^2\,\zeta_{nt}\zeta_{n't}\,K(x,y).$$

In the present implementation one has $a = \frac{1}{8}$, which gives the effective coefficient -4 derived above. This is the only remaining universal scalar prefactor in the current construction.

However, the benchmark set shows that changing only this universal prefactor cannot by itself close the problem. A uniform rescaling of the off-diagonal uv block by factors 1/2, 1/4, or 1/8 lowers all species together, but OCS remains far too large throughout. So the residual discrepancy cannot be explained by a single missing numerical factor in front of the whole off-diagonal $U_{nn'}/V_{nn'}$ sector.

What this implies for the final observable. Within the final state dependence actually used in the working branch,

$$D_v = D_J - \sum_n \beta_n\left(v_n + \frac{1}{2}\right) - \sum_t \beta_t(v_t + 1),$$

there is no other comparably large parallel contribution available to cancel the off-diagonal uv channel once it has been formed. In all four benchmarks the quartic seed, quartic-r, mixed-prefactor, and X/F blocks are much smaller than the dominant off-diagonal uv piece whenever the parallel sector fails. Therefore, if a compensating mechanism exists in the exact linear formalism, it is not hidden among the already implemented terms with a slightly different prefactor. Moreover, a separate linear pair term of the form $-\gamma_{nn'}(v_n + v_{n'} + 1)$ would be strictly equivalent to shifting β_n and $\beta_{n'}$ by the same amount, so the missing ingredient cannot be merely an omitted observable term written in a different notation. It must correspond to a different analytic reduction of the off-diagonal $U_{nn'}/V_{nn'}$ sector before it enters the final mode coefficients. This is the sharpest statement that can be made from the present implementation and benchmark set.

What OCS shows. The addition of OCS makes the situation sharper. In OCS, the parallel sector is again dominated almost entirely by a single off-diagonal channel, now

$$n = 0 \longleftrightarrow n' = 1,$$

and the corresponding contribution is much larger:

$$uv_{01} \approx -2.743 \times 10^{-6}\ \mathrm{cm}^{-1}.$$

Numerically, the current working branch gives

$$\beta_\parallel \approx (-0.0822278787,\ -0.0822278766)\ \mathrm{MHz}, \qquad \beta_\perp \approx -1.1147101 \times 10^{-9}\ \mathrm{MHz}.$$

As in C_2H_2, the $r_{nn'}$ piece cancels and the surviving term is again a pure $B^2\zeta\zeta$ operator contribution. The OCS case therefore rules out the idea that the remaining parallel discrepancy could be fixed by a universal small normalization factor on `pair_offdiag`. The residual problem is systematic and belongs to the off-diagonal operator sector

$$U_{nn'}/V_{nn'} \quad (n \neq n')$$

itself.

Why HCN and HCCD do not show the same failure. The benchmark set also shows that the large residual is not universal across all linear molecules. The raw 2×2 pair blocks of HCN and HCCD are numerically close to

$$\begin{pmatrix} a & s \\ -s & a \end{pmatrix},$$

that is, a trace part plus an antisymmetric part. By contrast, the problematic blocks in C_2H_2 and OCS are close to

$$\begin{pmatrix} a & s \\ s & -a \end{pmatrix},$$

that is, symmetric and nearly traceless.

This matters because the working operator reduction `pair_offdiag` is exactly the symmetric off-diagonal component

$$\frac{M_{12} + M_{21}}{2}.$$

It therefore vanishes automatically on a purely antisymmetric block and is nonzero only when the degenerate pair block has genuine symmetric off-diagonal content. This is why HCN and HCCD do not exhibit the same large off-diagonal parallel uv channel, while C_2H_2C2H2 and OCS do.

This comparison also shows that the unresolved issue is not solved by simply switching from the off-diagonal symmetric component to the diagonal traceless one. In the same working machinery, replacing `pair_offdiag` by `pair_diag` lowers the OCS parallel residual strongly, but it does not close C_2H_2 and it drives HCN/HCCD into a clearly nonphysical regime. Therefore the remaining ambiguity is not "which one of the two canonical symmetric components is universally correct." The physically relevant Aliev component must correspond to a species- and pair-dependent direction inside the two-dimensional spin-2 space spanned by the traceless diagonal and symmetric off-diagonal parts of the pair block.

Connection to the Aliev pair basis. The present implementation does not choose the degenerate pair basis from the Gaussian ordering of modes. It reconstructs the pair basis from the first-level geometry tensor

$$c_1(i; ab),$$

itself reconstructed from $\mu_1 = d(I^{-1})/dQ$.

Setup. Let W_t be the two-dimensional harmonic eigenspace associated with one degenerate perpendicular pair. At the harmonic level, any orthonormal basis of W_t is admissible, so the pair has an internal gauge freedom

$$W_t \mapsto R\, W_t, \qquad R \in O(2).$$

Therefore the Gaussian ordering of the pair is not physical. To obtain a physical basis one needs an additional geometrical tensor that lifts this degeneracy.

For a linear molecule with symmetry axis s and perpendicular plane $\Pi_\perp = \mathrm{span}(p, q)$, the mode-resolved first rotational-derivative tensor induces, for each mode i, the four-component row

$$f_i = \big(c_1(i; s, p),\ c_1(i; s, q),\ c_1(i; p, p) - c_1(i; q, q),\ 2c_1(i; p, q)\big).$$

The first two entries are the vector part coupling the symmetry axis to $\Pi_\perp$; the last two entries are the symmetric traceless tensor part inside $\Pi_\perp$. This is exactly the irreducible geometry that distinguishes the linear-axis direction from the perpendicular plane at first order in the normal coordinates.

Proposition. Assume that for a degenerate pair (i, j) the geometry carried by c_1 is not identically isotropic on the pair subspace, i.e. the associated feature rows are not both zero and are not exactly equivalent up to the nongeneric fully isotropic case. Equivalently, assume that the two eigenvalues of the c_1-induced Gram matrix

$$G_{ij} = F_{ij}F_{ij}^T, \qquad F_{ij} = \begin{pmatrix} f_i \\ f_j \end{pmatrix},$$

are distinct:

$$\lambda_1 \neq \lambda_2.$$

Then G_{ij} selects a unique orthonormal basis of the pair subspace W_t, up to independent sign flips of the two basis vectors.

Proof. Under any orthogonal change of basis $R \in O(2)$ inside the degenerate pair, the row matrix transforms as

$$F_{ij} \mapsto RF_{ij}.$$

Therefore

$$G_{ij} \mapsto (RF_{ij})(RF_{ij})^T = R(F_{ij}F_{ij}^T)R^T = RG_{ij}R^T.$$

So G_{ij} is the intrinsic symmetric bilinear form induced by the geometry tensor c_1 on the pair subspace. Since G_{ij} is real symmetric, it admits an orthonormal eigenbasis. If its two eigenvalues are distinct, that eigenbasis is unique up to sign of each eigenvector. If the eigenvalues coincide, the geometry is isotropic on the pair and no preferred physical basis exists. In the present linear benchmarks the non-isotropic case holds, so the eigenbasis of G_{ij} is the unique physical basis selected by the first-order geometry. □

Consequence. Aliev's symbols (t_a, t_b) are not Gaussian labels; they denote a physical orthonormal basis of the degenerate perpendicular pair. In the present implementation, the only basis with that status is the eigenbasis of the c_1-induced Gram matrix G_{ij}. Therefore

$$\text{Aliev } (t_a, t_b) \equiv \text{ canonical pair basis selected from } c_1 \quad \text{(up to basis-vector signs).}$$

Once this basis is fixed, the operator component

$$\zeta^x_{n,tb}$$

is represented in the canonical pair block by the off-diagonal symmetric entry,

$$\texttt{pair\_offdiag} = \frac{M_{12} + M_{21}}{2}.$$

So the present bridge is not heuristic: it is the unique component extraction associated with the unique geometry-selected pair basis. With this identification, the operator reduction `pair_offdiag` is the component form compatible with the Aliev notation, while `principal_direction` remains excluded because it returns the norm of the degenerate block rather than one of its physical components.

11. Conclusions and status

The current linear branch should be read in the following way.

- The algebraic implementation of the linear families, of β_n, β_t, and of the optical scalar L is now explicit and internally consistent.

- The Gaussian bootstrap is no longer using the Gaussian Coriolis tensor as if it were already the physical Aliev ζ object.
- The general-to-linear reduction of the observable branch is now closed at the structural level: the final linear observable separates into a mode-anchored sector and a pair-anchored off-diagonal parallel sector.
- The recommended final implementation is therefore the decompacted linear observable, not the compact scalar 1986 branch. The compact branch remains as a legacy shorthand that is observationally legitimate only when the pair sector is negligible against the mode sector.
- Two issues remain if one insists on a final literature-validated compact branch:
  - the sign of the perpendicular $xf/cross$ insertion is still adopted as a working convention;
  - the compact 1986 off-diagonal parallel branch is not quantitatively sufficient as a universal final observable representation.

So the correct conclusion is:

> The linear-molecule branch is now closed in its general form. The final structurally correct implementation is the decompacted observable derived from the Aliev–Watson general formalism and reduced to the linear xy block. The compact 1986 formulas survive only as a legacy compressed branch, acceptable when the pair-anchored parallel sector is negligible. What remains open is not the general derivation, but only the possibility of recovering from that general branch a universally valid compact observable off-diagonal formula. The present work resolves this point negatively: no benchmark-independent universal compactification is supported by the general reduction or by the benchmark set.

Equivalently, the correct status of the present note is:

> rigorous implementation audit + structurally derived final general linear branch,

while the compact scalar 1986 branch remains only as a conditional legacy compression of that general branch.

12. Implementation policy

For practical use, the present note supports the following implementation policy.

- **Observable branch.** The default linear observable should be the decompacted general branch, in which mode-anchored and pair-anchored contributions are kept distinct.
- **Compact branch.** The compact 1986 scalar branch should be treated as a legacy compression and used only under an explicit compactness gate, i.e. only when the pair-anchored parallel sector is negligible compared with the mode-anchored one.
- **Quartic input.** The quartic sector should be implemented once at the full four-index level. Reduced three-index quartic data should enter only through an input adapter that populates the representable four-index entries and sets the non-representable ones to zero.

This policy is the cleanest way to keep the code aligned with the derivation: one general observable implementation, one legacy compact shorthand, and one general quartic backend with multiple input layers.